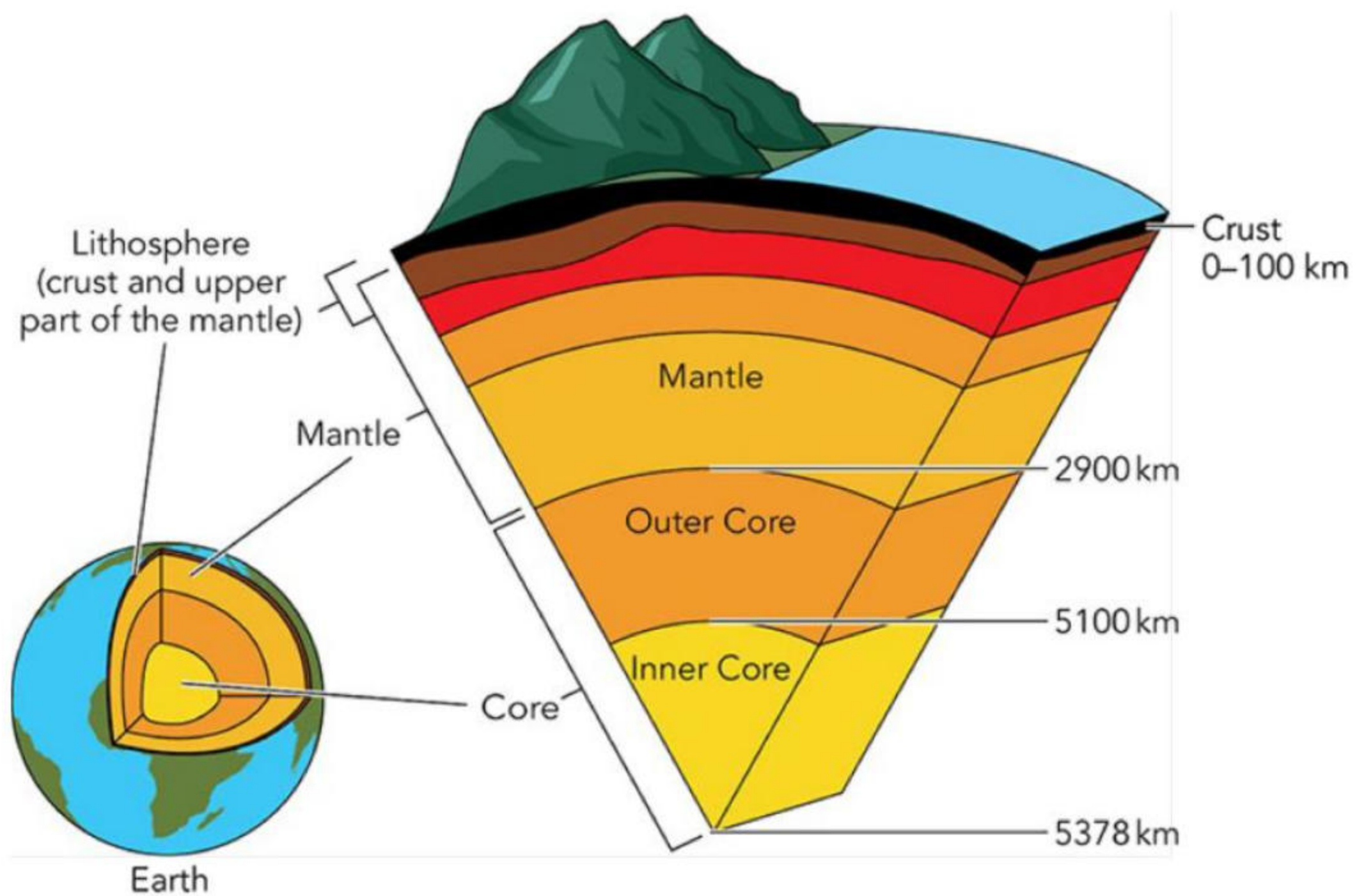


4.1 The Earth's structure

The Earth under our feet is far more complex than it looks (Source A). Over the past 200 years, scientists have used a number of investigative tools including satellites, photography, mapping of the oceans, and studying earthquakes to understand what the surface and the internal structure of the planet looks like.

The outer layer of the Earth is the **crust** (on average 50 km deep). The crust is solid rock and made up of a collection of **tectonic plates**. These plates join to form the outer shell like a puzzle. Beneath the crust is the **mantle** (estimated to be 3000 km deep), which is a layer of semi-molten rock called **magma**. Semi-molten means that the rock is in a state between liquid and solid. The outer layer of the mantle is the **lithosphere**. The lithosphere is made up of the crust and the upper part of the mantle. Beneath the mantle is the Earth's **core**. The core is divided into the outer core (estimated to be 2300 km deep) and the inner core (estimated to be 2400 km wide). The inner core is made up of mainly iron and nickel. It is the hottest zone in the Earth's structure, with a temperature of approximately 5200°C.



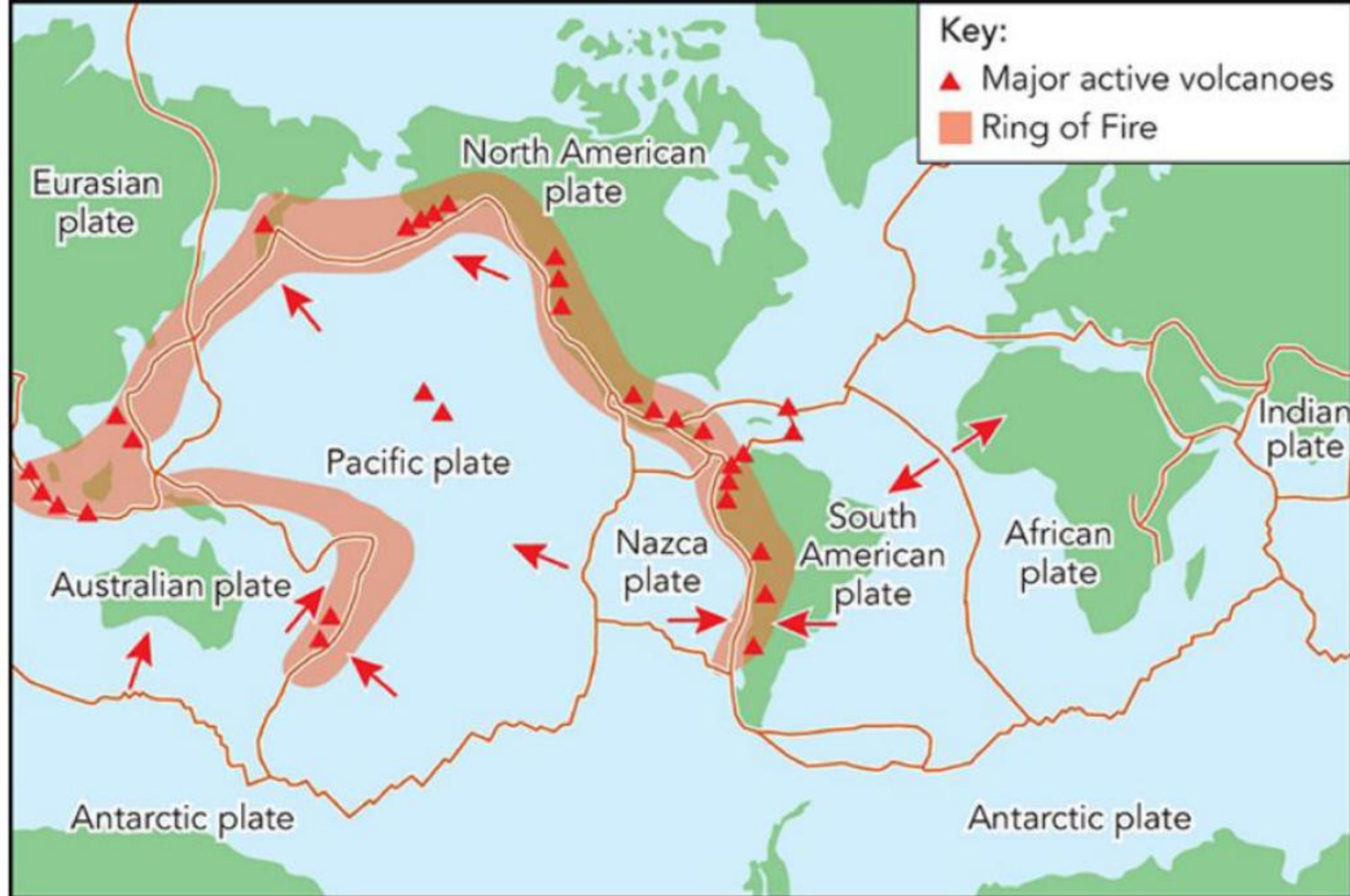
Source A: The structure of the Earth

Tectonic plates

Within the mantle, **convection currents** flow. Hotter material (rock and magma) rises and cooler material sinks. There are seven major tectonic plates that sit on top of this slow-moving magma. The force of convection currents drags the plates apart, pushes them together, or forces the plates to move past each other (Source B). A plate boundary is where these tectonic plates meet. It is at these plate boundaries we find most volcanoes and where most earthquakes occur. This movement also results in the formation of landforms such as mountain ranges, volcanoes, and oceanic trenches.

TASK 1: Study Source A

- Complete a labelled diagram** of the Earth's structure.
- State which layer of the Earth is the thinnest.
- In pairs, discuss how convection currents move the tectonic plates across the surface of the Earth.
- Draw a simple diagram** to show the movement of magma and the convection currents. Include the direction that the plates are moving in your diagram.



Source B: The plates that make up the Earth's crust

There are two main types of tectonic plate: oceanic plates and continental plates. Oceanic plates are younger and denser and found beneath the oceans. Continental plates are older and less dense and form the land masses that rise above the surface of the oceans. There are four main types of plate boundaries that form (Sources C–F), depending on how the Earth's crust is moving at any given point.

TASK 2: Study Source B

- Which plate do you live on? Consider your distance from the nearest plate boundary. Are you at risk from earthquakes or volcanoes?
- Describe the direction of plate movement between the Pacific plate and the Nazca plate.
- Name two plates that are moving towards each other. What do you call this type of plate boundary?
- In pairs, discuss the shape of the continents. Consider how the continents could fit together if you were to cut them out to use as a puzzle. Is there a clear pattern? Does it suggest where continents may have been joined together in the past?

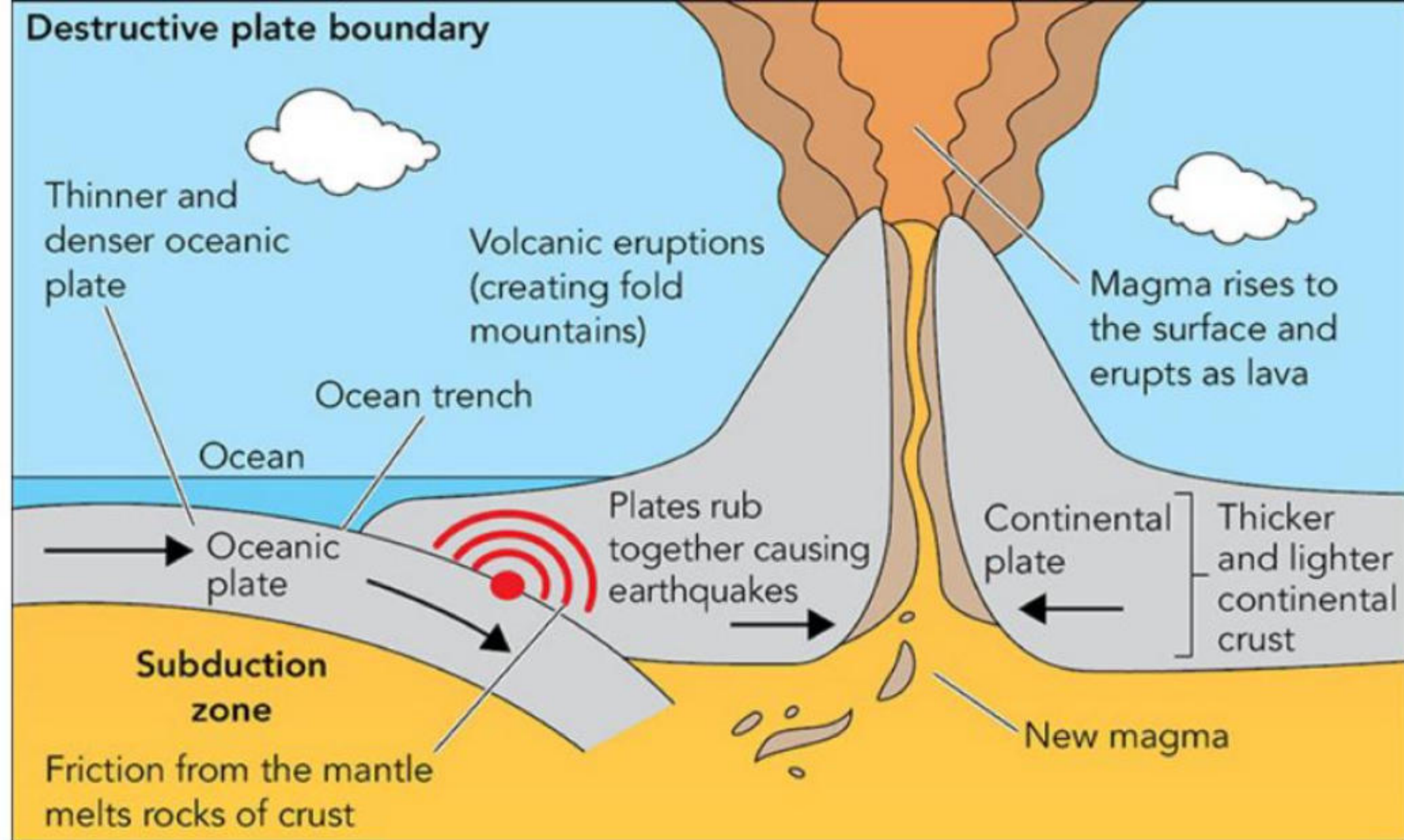
Convergent plate boundary

Convergent plate boundaries occur when two tectonic plates move towards each other. This can take two forms: a destructive plate boundary or a continental collision plate boundary.

Convergent and destructive plate boundary

A converging and destructive plate boundary is a complex plate boundary where two or more tectonic plates collide, and one plate **subducts** beneath another. Subduction is when one plate is forced beneath the other due to the difference in density. The younger, denser plate sinks beneath the older, less dense plate. This type of boundary forms between a continental plate and an oceanic plate, or an oceanic plate and another oceanic plate.

The subducting plate melts due to friction and pressure. This forms a magma chamber beneath the surface. Pressure then forces the magma chamber to the surface and it creates a **volcano**. At the site of the collision, where the two plates meet, a deep ocean trench is formed. On the surface, the plate that is forced over the top of the subducting plate buckles to form a fold mountain range. An example is the plate boundary between the Pacific plate and the Nazca plate (South America).

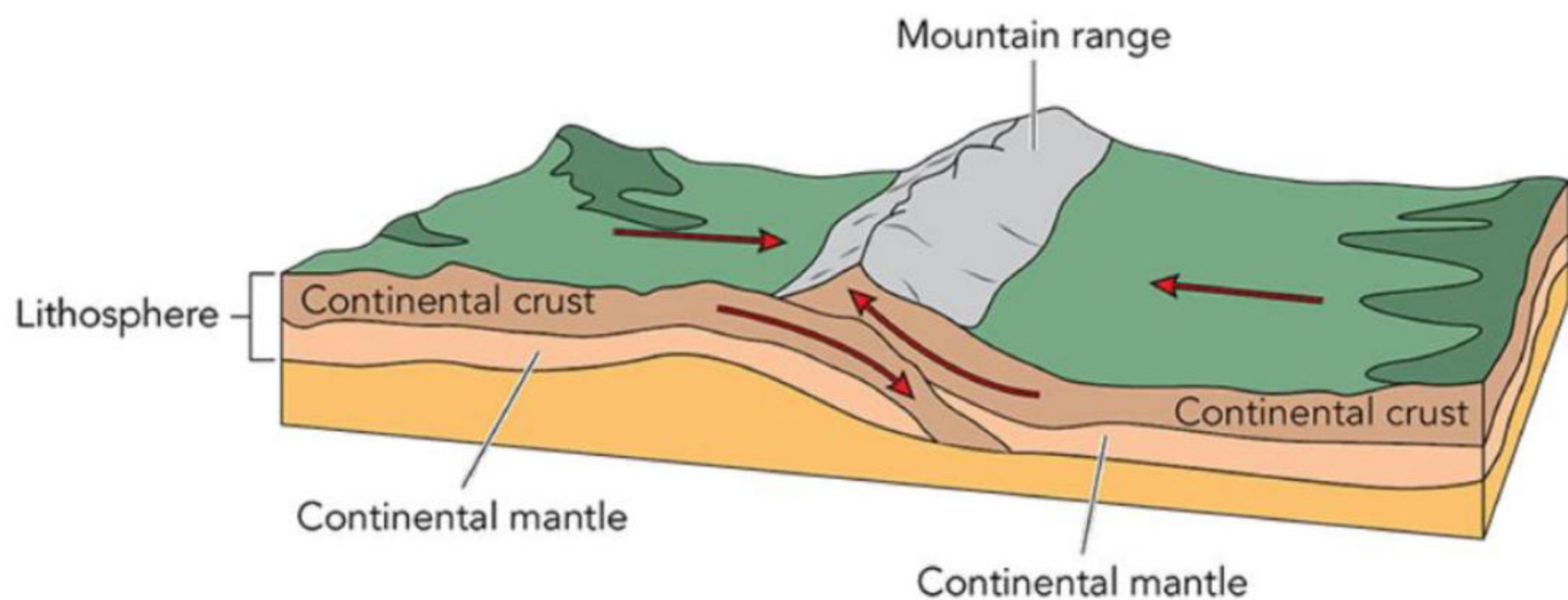


Source **C**: A convergent/destructive plate boundary

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Convergent and continental collision plate boundary

A converging collision plate boundary is where two continental plates collide. These plates are of equal density, so neither plate subducts. As a result, both plates crumple and fold, forming mountain ranges. An example of this is where the Indian sub-continent is moving north into the Asian plate. This formed and continues to form the Himalayan Mountain range. These plate boundaries experience earthquakes as the plates move towards each other.



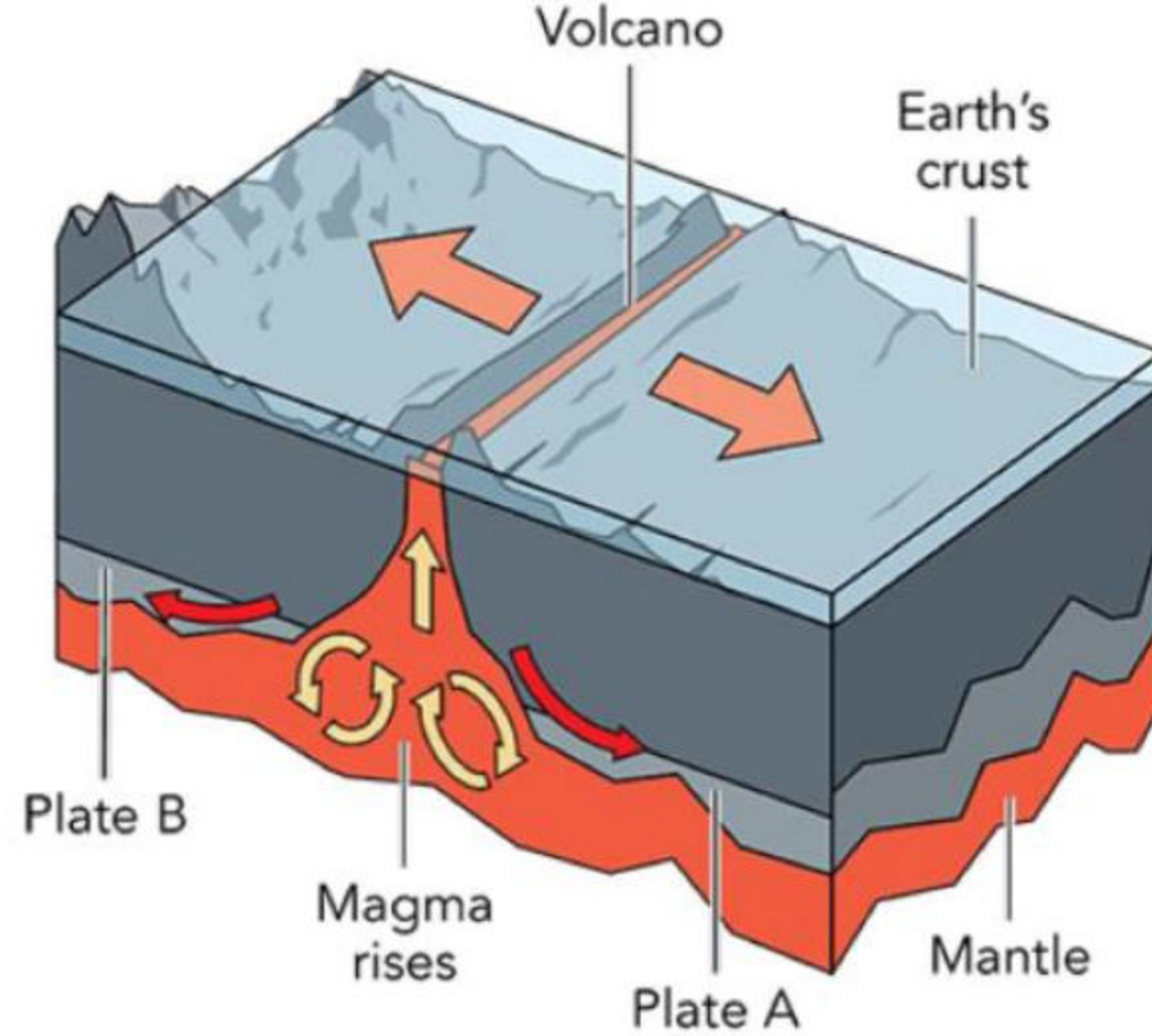
Source **D**: A convergent/continental collision plate boundary

Divergent or constructive plate boundary

Divergent (or constructive) plate boundaries occur where two plates are moving away from each other. This forms a gap that fills with magma rising from the mantle. As the magma cools, this creates (or 'constructs') new crust.

This type of plate boundary is found on both oceanic and continental plates. An oceanic example is the Mid-Atlantic Ridge that runs up the middle of the Atlantic Ocean. The rising magma forms ocean ridges along both sides of the plate boundary. The ridges form submarine mountain ranges.

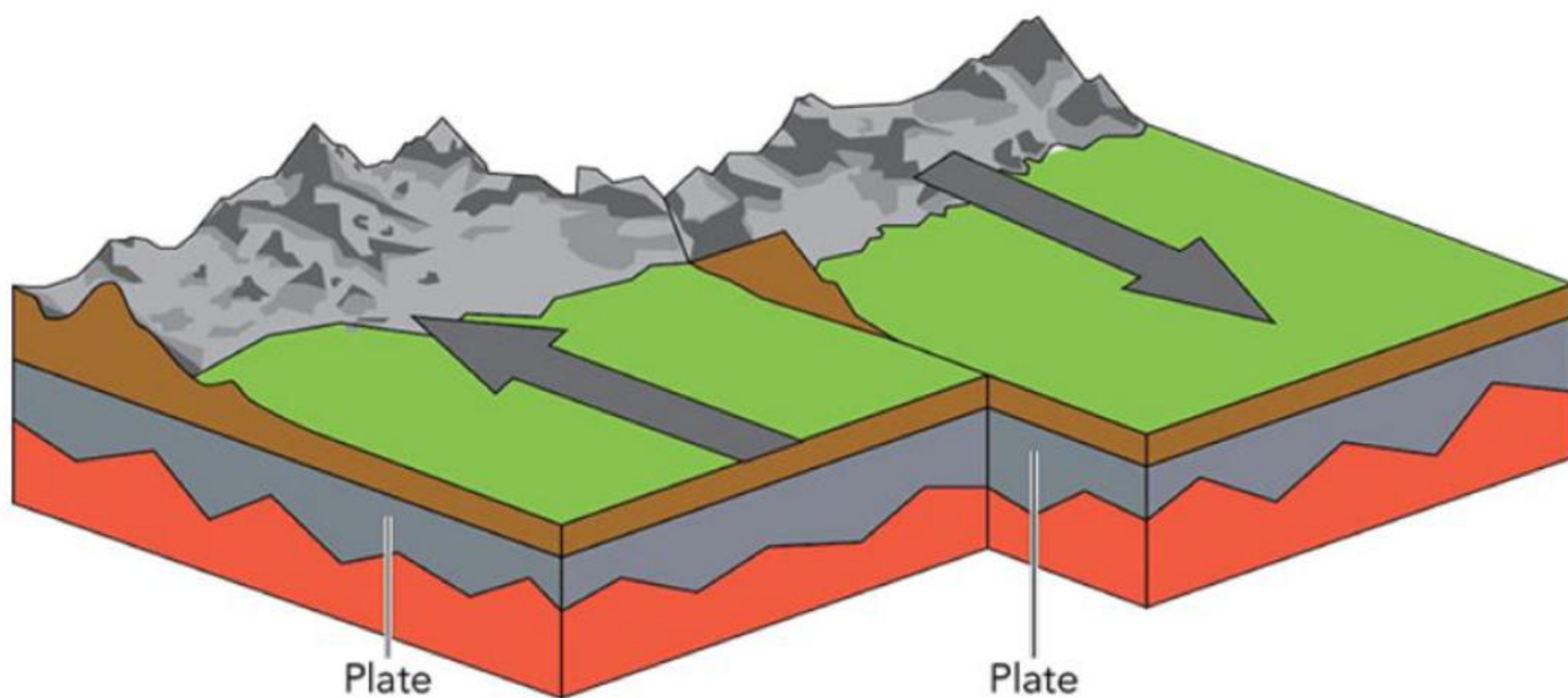
A continental example is the Great Rift Valley in Africa. The Great Rift Valley forms where the continent is splitting apart.



Source **E**: Divergent/constructive plate boundary

Conservative or transform plate boundary

Conservative (or transform) plate boundaries occur where two plates move past each other in the same direction at different speeds, or in opposite directions. The plates do not move smoothly past each other, often building up friction. As the plates slip, **earthquakes** occur.



Source **F**: Conservative/transform plate boundary

TASK 3: Study Sources **B C D E F**

- With reference to the diagrams, **complete a clearly labelled sketch diagram** of a convergent/destructive plate boundary.
- With reference to Source B and your understanding of plate boundaries, explain what type of plate boundary is found around the Ring of Fire.
- Working in pairs, use the template below to create a table that compares the differences between the plate boundaries. Use the diagrams and the information on plate boundaries to identify different characteristics and then list six characteristics that you can compare. Fill in the ticks to show which plate boundary has that characteristic. Two examples have been provided for you.

Type of plate boundary	Divergent/Constructive	Convergent/Destructive	Convergent/Collision	Conservative
Land is being formed	✓	✗	✗	✗
Land is being destroyed	✗	✓	✗	✗

REFLECTION

When you created the plate boundary table in Task 3, how difficult did you find it to compare plate boundaries? Did working in pairs help you to think of things you may otherwise not have noticed?

